

BOA VISTA, Brazil

WMO: 820220

Lat: 2.846N

Lon: 60.690W

Elev: 276

StdP: 14.55

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	71.7	73.1	59.0	75.3	90.5	60.8	80.4	90.1	18.8	80.7	15.9	83.1	5.2	60	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	16.7	96.7	76.1	95.3	75.8	94.8	75.6	79.1	89.4	78.7	88.8	78.2	88.0	9.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	142.8	81.4	76.5	139.8	81.1	75.6	135.3	80.6	42.7	89.6	42.2	89.6	41.7	88.4	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.4	18.8	16.7	69.6	98.8	1.9	1.9	68.2	100.1	67.0	101.2	66.0	102.3	64.5	103.6		
			65.5	81.9	3.0	2.2	63.3	83.4	61.5	84.7	59.8	85.9	57.6	87.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.4	83.6	83.9	84.7	83.9	81.9	81.3	81.1	82.7	84.5	85.1	84.8	83.9	(6)
(7)		DBStd	2.73	1.88	1.95	2.24	2.74	2.81	2.27	2.28	2.55	2.41	2.60	2.53	2.18	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12207	1043	949	1077	1018	989	938	964	1014	1034	1088	1043	1050	(10)
(11)		CDD65	6732	578	529	612	568	524	488	499	549	584	623	593	585	(11)
(12)		CDH74	76420	6795	6328	7674	6676	5259	4552	4512	5699	6966	7765	7314	6880	(12)
(13)		CDH80	33668	3046	2885	3713	3015	1915	1504	1473	2298	3294	3865	3554	3106	(13)
(14)	Wind	WSAvg	7.4	10.6	11.0	10.7	8.4	6.3	5.1	4.7	4.8	5.6	6.4	6.8	8.7	(14)
(15)	Precipitation	PrecAvg	60.9	0.9	0.7	2.0	5.6	10.4	12.5	11.5	7.8	3.9	2.3	2.6	1.7	(15)
(16)		PrecMax	87.1	3.9	3.0	11.7	18.0	22.2	25.6	21.9	14.6	9.9	8.3	10.6	9.2	(16)
(17)		PrecMin	9.0	0.0	0.0	0.0	0.0	0.7	0.9	0.7	0.4	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	18.0	1.0	0.7	2.8	4.4	5.2	5.7	4.8	3.4	2.8	2.0	2.3	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.4	96.5	97.1	96.7	94.1	92.8	92.6	95.3	96.9	97.2	97.1	96.4	(19)
(20)			MCWB	75.3	75.5	74.4	74.7	76.0	77.3	76.7	77.0	77.3	76.4	77.3	76.5	(20)
(21)		2%	DB	93.5	93.6	95.2	94.9	92.5	91.0	90.6	93.4	95.2	96.3	95.3	94.5	(21)
(22)			MCWB	74.3	73.7	74.3	74.7	75.8	76.7	76.3	76.8	76.5	76.2	76.5	76.1	(22)
(23)		5%	DB	92.4	92.8	93.6	93.2	90.5	89.4	89.3	91.7	93.6	94.9	94.3	92.9	(23)
(24)			MCWB	74.0	73.5	73.8	74.5	76.1	76.5	76.1	76.4	76.0	75.9	76.0	75.5	(24)
(25)		10%	DB	91.2	91.3	92.7	91.5	88.6	87.6	87.5	89.9	92.8	93.3	93.0	91.3	(25)
(26)			MCWB	73.9	73.6	73.7	74.8	76.2	76.3	76.2	76.4	75.9	75.5	75.4	74.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.1	78.3	78.5	79.1	79.5	79.1	78.7	79.4	79.2	79.2	79.3	79.1	(27)
(28)			MCDWB	89.5	89.0	89.5	86.9	87.4	86.5	87.0	89.6	90.7	91.9	91.8	90.4	(28)
(29)		2%	WB	76.9	76.9	77.2	78.1	78.6	78.2	77.9	78.5	78.5	78.3	78.3	78.0	(29)
(30)			MCDWB	88.3	88.3	87.9	86.3	86.1	85.7	85.7	88.2	89.8	90.3	90.5	88.8	(30)
(31)		5%	WB	76.2	76.1	76.3	77.3	77.9	77.6	77.3	77.9	77.9	77.8	77.6	77.2	(31)
(32)			MCDWB	87.2	87.2	87.1	85.7	85.1	84.8	84.7	86.9	88.6	89.2	88.8	87.4	(32)
(33)		10%	WB	75.5	75.4	75.5	76.7	77.2	77.1	76.7	77.2	77.3	77.2	77.0	76.4	(33)
(34)			MCDWB	86.1	86.2	86.4	85.2	84.2	84.0	83.7	85.7	87.3	88.0	87.4	86.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.6	15.7	15.8	14.7	12.8	12.8	13.1	14.7	16.3	16.7	16.3	15.5	(35)
(36)			MCDBR	17.0	17.3	17.3	16.8	15.4	14.9	15.1	16.7	17.5	17.9	17.7	17.1	(36)
(37)			MCWBR	4.1	4.3	4.1	3.9	4.2	4.3	4.2	4.4	4.2	3.9	3.9	3.9	(37)
(38)		5% WB	MCDWB	15.2	15.2	15.4	14.4	13.3	13.3	13.6	15.3	16.6	16.9	16.5	15.4	(38)
(39)			MCWBR	4.5	4.8	4.6	4.4	4.5	4.5	4.5	4.7	4.6	4.5	4.4	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.390	0.404	0.409	0.399	0.393	0.397	0.407	0.422	0.433	0.422	0.418	0.407	(40)	
(41)		taud	2.516	2.477	2.469	2.512	2.527	2.526	2.488	2.422	2.373	2.414	2.425	2.464	(41)	
(42)		Ebn at Noon	296	293	290	287	282	277	276	277	279	285	285	289	(42)	
(43)		Edh at Noon	35	37	37	35	33	32	34	37	40	38	38	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1658	1653	1693	1587	1416	1473	1509	1694	1871	1826	1703	1579	(44)	
(45)		RadStd	89	95	95	107	125	102	105	108	86	72	93	82	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) -0.76	(e) -0.59	(f) -2.61	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page